Exp No:2 Upload and Analyze the data set given in csv format and perform data.

Aim:

To Upload and Analyze the data set given in csv format and perform data.

Description: Use sample data set sales-data.csv.

Code:

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import io
df = pd.read_excel(io.BytesIO(uploaded['SALES DATA.xlsx']))
print(df.head())
print(df.isnull().sum())
df['Sales'].fillna(df['Sales'].mean(),inplace=True)
df.dropna(subset=['Product','Quantity','Region'],inplace=True)
print(df.describe)


product_summary=df.groupby('Product').agg(
  {
    'Sales':'sum',
    'Quantity':'sum'
  }
).reset_index()
print(product_summary)
plt.figure(figsize=(10,6))
plt.bar(product_summary['Product'],product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales By Product')
plt.show()
```

```
df['Date'] = pd.to_datetime(df['Date'])

sales_over_time = df.groupby('Date').agg({'Sales':'sum'}).reset_index()

plt.figure(figsize=(10,6))

plt.plot(sales_over_time['Date'],sales_over_time['Sales'])

plt.xlabel('Date')

plt.ylabel('Total Sales')

plt.title('Sales Over Time')

plt.show()


pivot_table=df.pivot_table(values='Sales',index='Region',columns='Product',aggfunc=np.sum,fill_va
lue=0)

print(pivot_table)


correlation_matrix=df.corr(numeric_only=True)

print(correlation_matrix)

import seaborn as sns

plt.figure(figsize=(8,6))

sns.heatmap(correlation_matrix,annot=True,cmap='coolwarm')

plt.title('Correlation Matrix')

plt.show()
```

Output:

	Date	Product	Sales	Quantity	Region
0	2023-01-01 00:00:00	Product A	200	4	North
1	2023-02-01 00:00:00	Product B	150	3	South
2	2023-03-01 00:00:00	Product A	220	5	North
3	2023-04-01 00:00:00	Product C	300	6	East
4	2023-05-01 00:00:00	Product B	180	4	West

Date 0

Product 0

Sales 0

Quantity 0

Region 0

dtype: int64

<bound method NDFrame.describe of Date Product Sales Quantity Region

0 2023-01-01 00:00:00 Product A 200 4 North

1 2023-02-01 00:00:00 Product B 150 3 South

2 2023-03-01 00:00:00 Product A 220 5 North

3 2023-04-01 00:00:00 Product C 300 6 East

4 2023-05-01 00:00:00 Product B 180 4 West

5 2023-06-01 00:00:00 Product A 210 5 North

6 2023-07-01 00:00:00 Product C 320 7 East

7 2023-08-01 00:00:00 Product B 160 3 South

8 2023-09-01 00:00:00 Product A 230 6 North

9 2023-10-01 00:00:00 Product C 310 7 East

10 2023-11-01 00:00:00 Product B 190 4 West

11 2023-12-01 00:00:00 Product A 240 6 North

12 13-01-2023 Product C 330 8 East

13 14-01-2023 Product B 170 3 South

14 15-01-2023 Product A 250 7 North

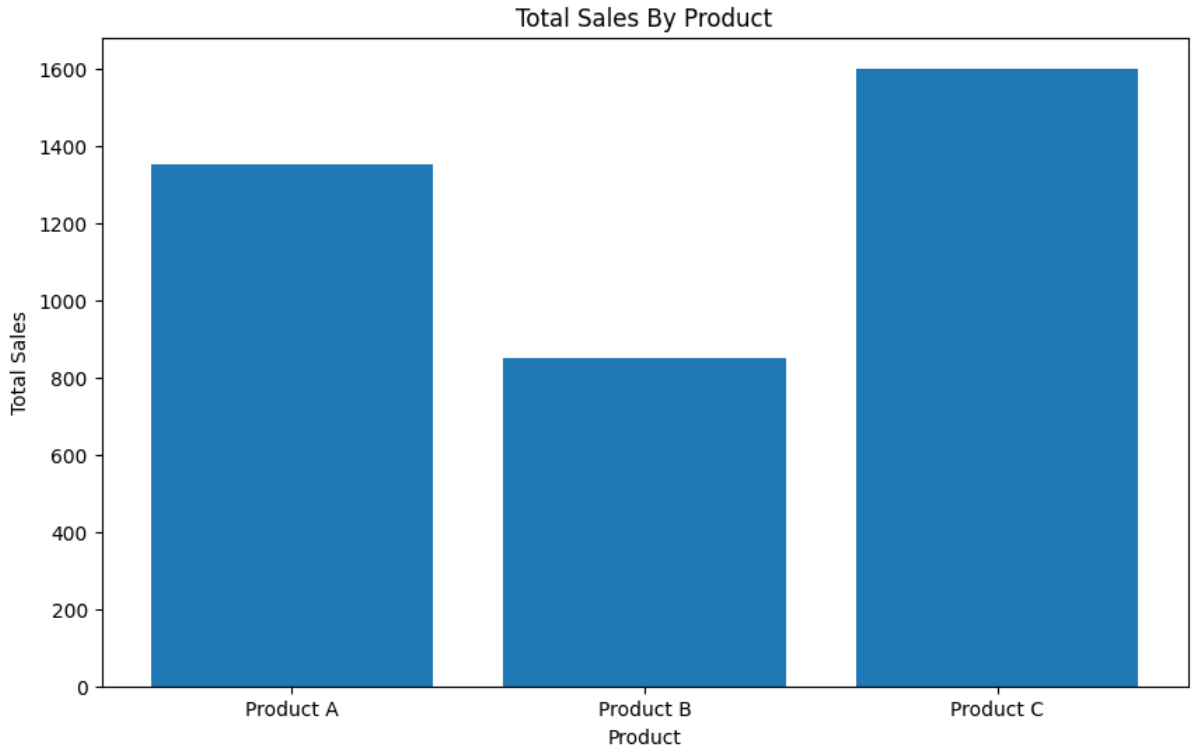
15 16-01-2023 Product C 340 8 East>

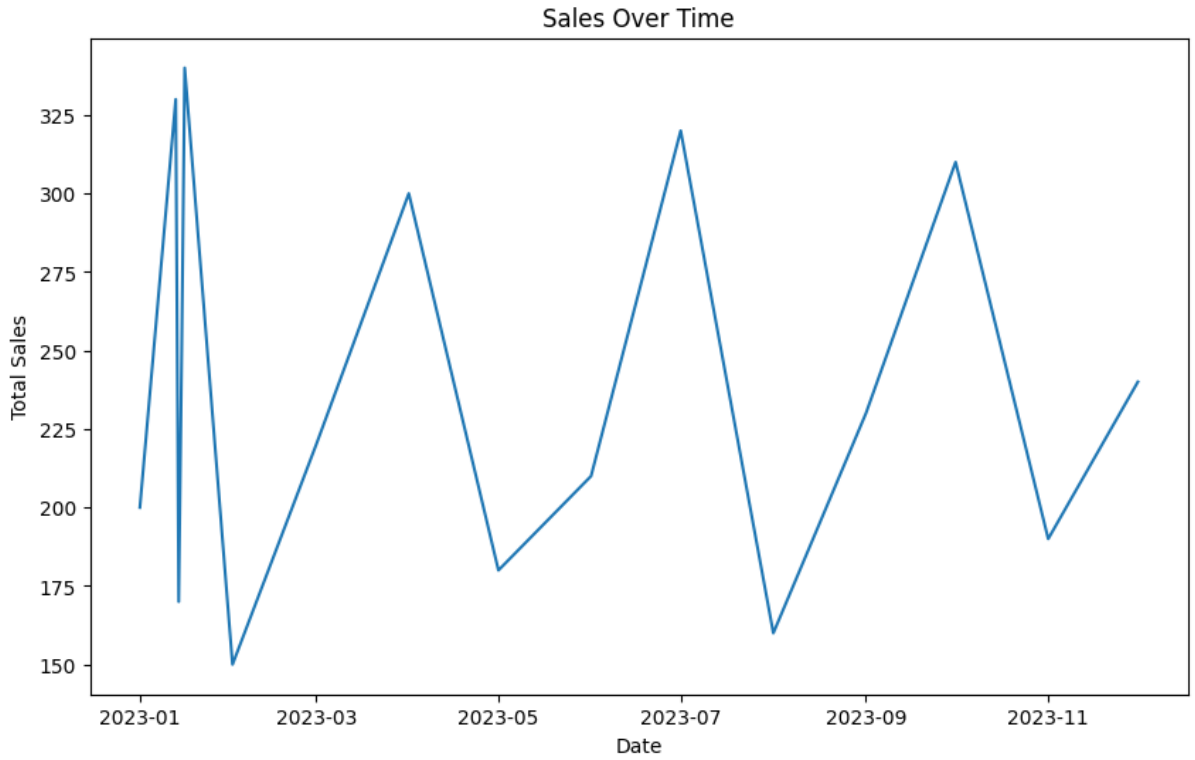
Product Sales Quantity

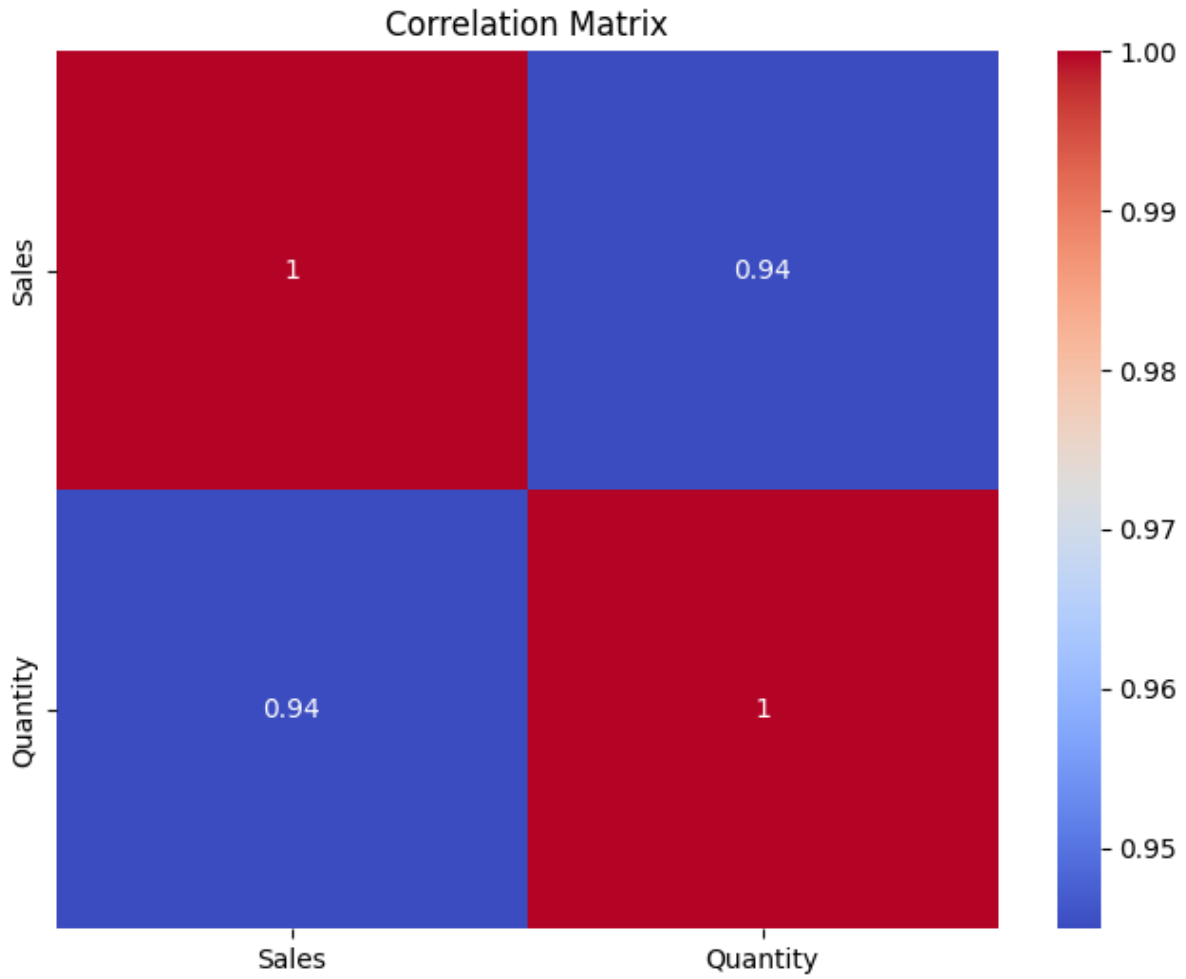
0 Product A 1350 33

1 Product B 850 17

2 Product C 1600 36


Total Sales By Product
Total Sales
1600
1400
1200
1000
800
600
400
200
0
Product A
Product B
Product C
Product


Sales Over Time
Total Sales
325
300
275
250
225
200
175
150
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
Date


Correlation Matrix
Sales
Quantity
1
0.94
0.94
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95